NLP PT1 Syllabus 2026

Chapter 1
Origin & History of NLP; Language, Knowledge and Grammar in language processing; Stages in NLP;Ambiguities and its types in English and Indian Regional Llanguages; Challenges of NLP;Applications of NLP

Chapter 2
Basic Terms: Tokenization, Stemming, Lemmatization; Survey of English Morphology, Inflectional Morphology, Derivational Morphology; Regular expression with types;

Morphological Models: Dictionary lookup, finite state morphology; Morphological parsing with FST (Finite State Transducer);Lexicon free FST Porter Stemmer algorithm; Grams and its variation: Bigram, Trigram; Simple (Unsmoothed) N-grams;

N-gram Sensitivity to the Training Corpus; Unknown Words: Open versus closed vocabulary tasks; Evaluating N-grams: Perplexity;
Smoothing: Laplace Smoothing, Good-Turing Discounting;

Part-Of-Speech tagging(POS); Tag set for English (Upenn Treebank); Difficulties /Challenges in POS tagging; Rule-based, Stochastic and Transformation-based tagging; Generative Model: Hidden Markov Model (HMM Viterbi) for POS tagging;

Chapter 3
Part-Of-Speech tagging(POS); Tag set for English (Upenn Treebank); Difficulties /Challenges in POS tagging; Rule-based, Stochastic and Transformation-based tagging; Generative Model: Hidden Markov Model (HMM Viterbi) for POS tagging;

NLP PT1 QUESTION BANK

1. What is Natural language processing? Explain generic NLP system?

2. What are the components of NLP?

3.Explain about the phases of Natural Language processing ?

4. Explain challenges and applications of NLP?

5.Explain types of ambiguities and ways to resolve it with examples?

6. What is the output of Morphological Analysis for Regular Verb, Irregular Verb, Singular noun, Plural noun.

7.What is the significance of FSA in morphological analysis ?

8.Explain Role of FST in Morphological Parsing with an example

9 Explain pre-processing steps generally used in NLP.

10.. Compare Stemming and lemmatization ?

11. Explain Porter Stemmer algorithm in detail

12. Illustrate Inflectional and derivational morphology with an example?

13. Explain Affixes and types of Affixes

14. What is a regular expression ? What is significance in word level analysis

15. Difference between free and bound morpheme ?

16. Explain following Terms :

- free and bound morpheme
- Morphotactics
- Orthographic rule

17. Explain Hidden Markov model for POS based tagging? What are the limitations of the HMM model?

18.Explain n-gram language model and its application ?

19. Numericals

Q1)Given a Hidden Markov model for POS tagging with initial probability, transition and emission probabilities . Compute **the most likely tag sequence** using Viterbi method

Sentence: (**the light book | DT JJ NN**)

$$\pi = Initial\ Probability = \{0.45,\ 0.29,\ 0.01,\ 0.25\}$$

Transition Probability Matrix (A)

	DT	NN	VB	JJ
DT	0	0.5	0.00001	0.60
NN	0	0.2	0.30	0.002
VB	0	0.3	0.10	0
JJ	0	0.2	0.001	0

Emission Probability Matrix (B)

	the	light	book
DT	0.30	0	0
NN	0.1	0.003	0.003
VB	0	0.06	0.01
JJ	0	0.002	0

Q2) Find the likelihood of the sequence "Book/VB Ticket/NN "

Transition Probability Matrix (A)

	NN	VB
<s>	0.7	0.3
NN	0.2	0.7
VB	0.7	0.2

Emission Probability Matrix (A)

	Book	Ticket
NN	0.4	0.6
VB	0.3	0.7

Q3) Consider the following corpus

<s> a/DT dog/NN chases/V a/DT cat/NN </s>
<s> the/DT dog/NN barks/V loudly/RB </s>
<s> a/DT cat/NN runs/V fast/RB </s>

i)Compute the emission and transition probabilities for a bigram HMM.
ii)Decode the following sentence using the Viterbi algorithm. "The cat chases the dog"

Q4) Given Corpus:

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<s> I am Sam </s>
<s> I like college </s>
<s> Do Sam like college </s>
<s> Sam I am </s>
<s> Do I like Sam </s>
<s> Do I like college </s>
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`<s> I do like Sam </s>`

List all possible bigrams. Compute conditional probabilities and predict the next word for the word

i) like ii) Sam

Which of the following sentence is better

1) <s> Sam I am</s>
2) <s> I do like Sam</s>